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Physics Internal Assessment

How is a swing's damping constant affected by the diameter (thickness) of its rope ?

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AIM : How is a swing's damping constant affected by the diameter (thickness) of its rope ?

1. INTRODUCTION :

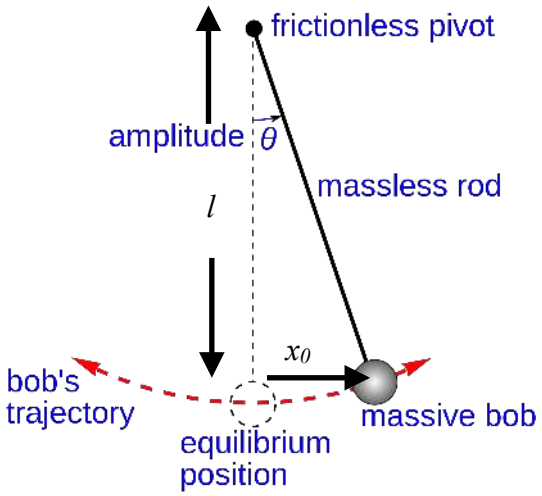
My myriad of interests have included pendulums for a long time. I enjoyed going to the beach as a child with my family and friends. Additionally, I usually spent a large deal of my time playing on the swing at the beach when we were there. There is no shame in admitting that, if given the opportunity, I would still play on the swing for hours even now. When I realised that a swing was simply a large pendulum with us acting as the bob, I at once developed a strange fondness for pendulums. I've been experimenting with pendulums ever since. In addition, I always made an effort to include pendulums into my high school projects. When my friends and I were at the beach one day, we saw two swings and decided to play on them for a bit. My friend and I went first while the rest pushed us from behind. My companion and I were both sent into the air to a comparable height following a strong push, however after a few seconds my swing ended while his continued. Because I had to give up the swing so quickly for the next person I was really bummed out. That day, on the way home, I tried to determine why my swing had abruptly halted. Were the bolts rusted? Or did my feet make contact with the earth to provide some resistance? Heavier than I, was he ? - No, we both were the same weight. After a few days, I came across an article by Mohazabbi and Shankar on damping[1]. The essay was based on an experiment the two of them had conducted to determine the impact of the string diameter on the pendulum's damping constant. As I thought about the swing again, this idea piqued my curiosity. Perhaps the swing came to an abrupt halt because the ropes on the two strings' thicknesses differed noticeably. Since I may now choose the topic of my IA, I want to investigate how the thickness of the rope in a swing affects the damping. However, measuring the damping of a swing is rather challenging so, I'll conduct my investigation using a simple pendulum. Thus formulating the below research question :

RESEARCH QUESTION : How is a pendulum's damping constant affected by the diameter (thickness) of its string ?

BACKGROUND INFORMATION :

WHAT IS A PENDULUM ?

A weight hanging from a pivot such that it may freely swing is called a **pendulum**[2]. A pendulum is subject to a restoring force from gravity when it is sideways moved from its resting, equilibrium position. This force accelerates the pendulum back toward the equilibrium position. When the pendulum is let go, the restoring force acting on its mass causes it to swing back and forth about its equilibrium point. The period is the length of time needed for one full cycle, or a left swing and a right swing. The period is influenced by the pendulum's length and, to a lesser extent, by the amplitude, or swing's width.



(Figure 1.1)

[1] *Damping of a Simple Pendulum Due to Drag on Its String*. 30 Dec. 2016, www.scirp.org/journal/paperinformation.aspx?paperid=73856

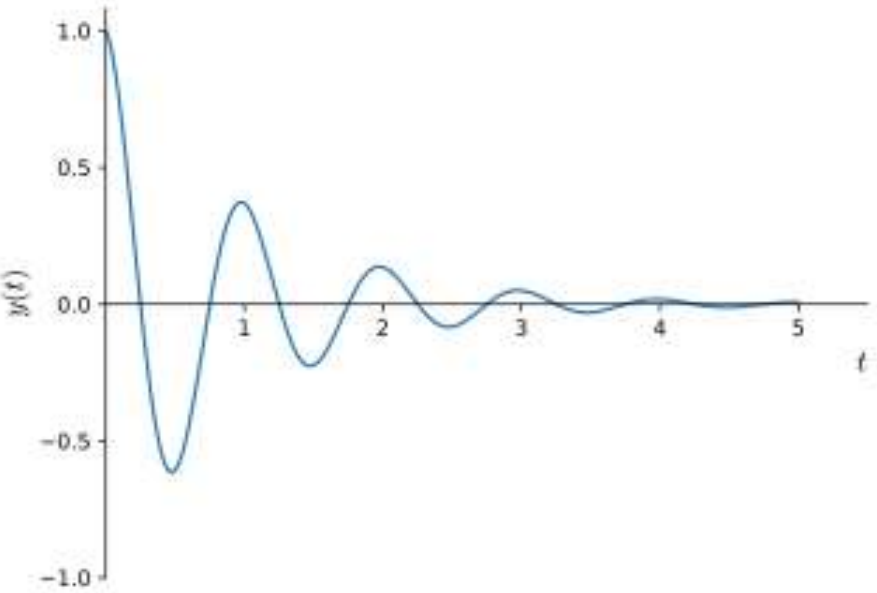
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Theoretically, once a pendulum is displaced, its oscillation must continue forever. Due to variables that oppose the oscillatory motion of the pendulum, this is not practically feasible. These elements might be the air's drag force or the resistance the pendulum faces at its suspension point. If there is no air resistance, the pendulum motion may be represented by a sinusoidal function of time. The pendulum motion is dampened by the frictional force brought on by air resistance.

WHAT IS DAMPING ?

A **damping** effect is one that reduces or stops oscillation in an oscillatory system by affecting it internally or externally[3]. In physical systems, processes that dissipate the energy contained in the oscillation cause damping.

Damping is experienced by most oscillatory systems such as harmonic oscillators. These systems, when displaced from their equilibrium position, experience a restoring force proportional to their displacement. If a frictional force proportional to the velocity is also applied, the system exhibits damped harmonic oscillation. The system's **damping coefficient** is a measure of how quickly it returns to rest as the frictional force dissipates its oscillation energy. There are several kinds of damping, but in this experiment we utilise mild damping, allowing the oscillations to continue but



with a gradually decreasing amplitude.

The displacement vs time graph of a pendulum when under damping effect (*Figure 2.1)*[4]

[3] "15.5 Damped Oscillations | University Physics Volume 1." *Courses.lumenlearning.com*, courses.lumenlearning.com/suny-osuniversityphysics/chapter/15-5-damped-oscillations/

[4]The Editors of Encyclopaedia Britannica. "Damping | Physics." *Encyclopædia Britannica*, 2019, www.britannica.com/science/damping.

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DERIVATION OF FORMULA :

Since a formula cannot directly relate the data that may be collected for the experiment. Therefore, derivation is necessary in order to analyse the collected data. We are aware that the formula for angular velocity (ω) is as follows :

$$\omega = \frac{v}{r} = \frac{v}{l} = \frac{2\pi}{T}$$

(Equation 1.1)

Where, *v* is the linear velocity of the pendulum (*ms*^-1) , *r* is the radius (*m*), *l* is the length (*m*) of the pendulum and *T* (*s*) is the time period of the pendulum. Thus the unit for the angular velocity (ω) is (*s*^-1). Since the radius of oscillation of the string is nothing but the length of the string, *r* has been substituted by *l* in the equation.

Since 2π in the above equation is the total angle of oscillation covered by the pendulum, this can be substituted by 4θ, as θ is the angular displacement (Figure 1.1) of the pendulum and it covers 4θ in one complete oscillation. Thereby giving the formula : $\omega = \frac{4\theta}{T}$

(Equation 1.2)

By equation "Equation 1.1" and "Equation 1.2", we get the equation : $\theta = \frac{vT}{4l}$

(Equation 1.3)

Since the amplitude of the oscillation decreases exponentially over time, the angle of displacement too decreases over time and can be written in the form :
$\theta = \theta_0 e^{-\frac{kt}{2}}$

(Equation 1.4)

where *k* is the damping constant of the pendulum.

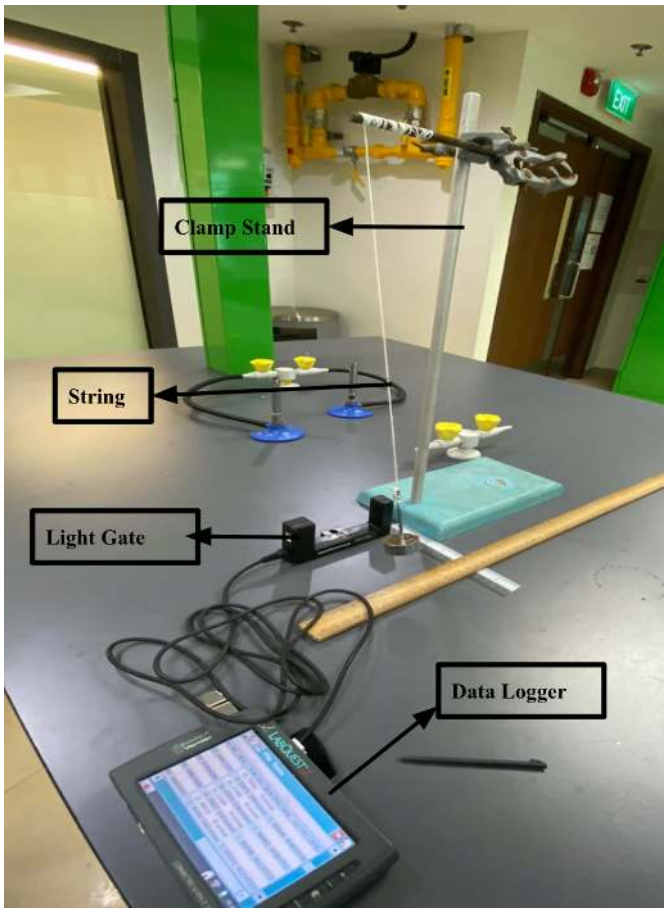
Taking "*ln*" on both sides : $ln(\theta) = ln(\theta_0 e^{-\frac{kt}{2}})$

(Equation 1.5)

We can derive the final equation as :
$\ln\left(\frac{\theta}{\theta_0}\right) = -\frac{kt}{2}$

(Equation 1.6)

The gradient from this last expression may be graphed to get the damping constant. The slope is $-\frac{k}{2}$, when $ln\frac{\theta}{\theta_0}$ is graphed against *t*. The damping constant, k, has a numerical value that is obtained by multiplying this number by -2.



Picture of the experimental setup(Figure 3.1)

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2. EXPLORATION

Hypothesis :The dampening effect of the pendulum changes as the string's diameter changes. This is because damping results in a decrease in velocity, and a decrease in amplitude is indicated by a decrease in velocity. As the string thickness increases, the velocity will decrease more quickly, resulting in a high string damping constant. The damping constant rises as the string thickness in the pendulum grows since it is directly proportional to string thickness.

Independent Variables :The length of the pendulum's string (680, 540, 480, 300, and 240μm, uncertainty = ±10μm). These string diameter values have been determined using a screw gauge.

Dependent Variables:The damping constant of the pendulum.

Controlled Variables :

Controlled Variables	Why is it important to control ?	How this experiment will control or monitor the variable ?
Mass of the pendulum bob	As changing the mass of the bob would change the pendulum's damping constant and lead to abnormalities in the experiment, it is crucial to control it.	By keeping the bob's mass constant and making sure it is the same bob that is used for the experiment with all the strings, we can regulate its mass.
Initial Displacement of the Pendulum	If the pendulums were randomly released from different points, anomalies would occur.	Place another clamp stand some distance away and release the bob from there. Demarcate the point where the bob was let go. Release all subsequent pendulums for all strings from the same point.
Material of the strings	Even if the variation brought on by the material of the string is essentially nonexistent, it is always preferable to avoid errors.	Use strings of the same material for the experiment.
Length of the string	The length of the string is vital in determining the damping of the pendulum. Thus it must be kept constant to avoid anomalies.	Stretch all the string together and cut them together using a definite length. For the experiment, I used strings of length 60cm. Incase there was excess length of a string, the excess was wound to the rod.
Force with which the pendulum was released	There would be differences in the pendulum's velocity when it swings through the light gate if it were released with varied initial forces; for example, if the pendulum was released with more force, the velocity of the pendulum would also rise.	To fix this issue, it was made sure that the pendulum was only released from one point and that no additional force was applied by having the same person release the pendulum for all the varied string diameters.

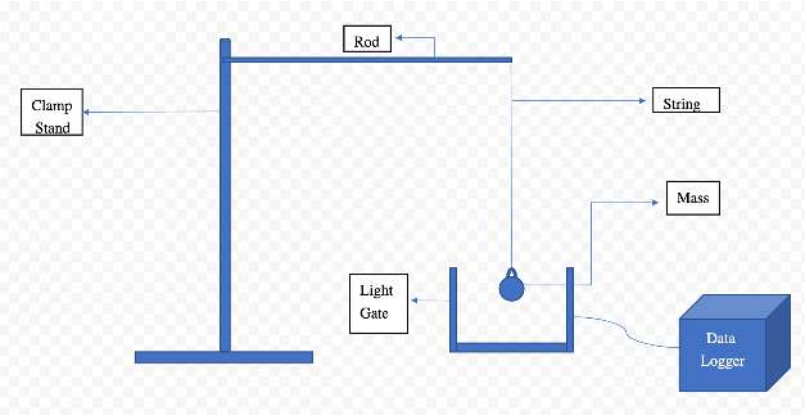
Control Variables (Table 1.1)

**Note : The damping constant of a pendulum is dependent on a variety of factors however, this experiment is only concerned about the effect of the thickness of the string on the damping constant. Therefore, the effect of the other factors on the damping constant must be made negligible. This can be done by keeping all these factors constant for all the strings used for this experiment.*

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Materials :

1. Strings of different diameters 680µm, 540µm, 480µm, 300µm, and 240µm (±10µm)
2. Clamp stand x 1
3. Vernier data logger x 1
4. Pendulum bob x 1
5. Light gate x 1
6. Rod x 1
7. Vernier Calliper (±10µm) x 1
8. Clamp x 1



Animated Experimental Setup(Figure 4.1)

Procedure :

1. Set up the experiment as shown in the diagram alongside (Figure 4.1). Hook the mass onto the string . To prevent movement in several directions, wind the string securely around the rod. Make sure the pendulum's bob is perfectly positioned between the light when it is at rest.
2. Measure 30cm from the setup and place a ruler at that point. Displace the pendulum to the ruler to release it.
3. Release the pendulum after starting the data logger. Make sure that the pendulum swings between the light gate and avoids hitting it.
4. Set the timer on the data logger for 100 seconds. Once 100 seconds have passed, the program on the data logger stops.
5. Transfer the data to a computer so that it can be analysed.
6. Repeat the procedures 1-5 using the other strings of different thicknesses.

Safety Concerns :

Concerns	Reason for concern	Method of control
The setup	The experiment is carried out on a table about 120cm from the ground, and if the apparatus were to fall, it may hurt our feet.	Place the setup away from the table's edge. Make sure that the arrangement is out of reach for children.
The pendulum	As the pendulum swings at a considerable height, there is a chance that it might strike you in the face when it is in the extreme position.	Ensure that you are at a safe distance away from the swinging bob.
The Pendulum bob	It is possible for the bob to become loose and separate from the pendulum as it swings back and forth. If the weight of the bob is too much it may affect the stability of the stand and cause it to fall.	Make sure the bob is correctly connected to the string and that its mass doesn't exceed what would cause the string to break and may effect the stability of the stand. Make sure that the stand is firmly fitted. Wear safety goggles to protect your eyes in case the pendulum breaks.

Safety Concerns (Table 2.1)

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Note : This experiment's ethical and environmental implications have been thoroughly examined, and none appear to exist.

3. ANALYSIS :

Raw Data :

Since it is not possible to fir in all the data I've collected for the five strings, given below(table 3.1), is the raw data collected from the first string used in this investigation of thickness 240µm.[5]

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)
2.3069	0.429664197	2.3101	0.428593175	2.3831	0.429828656	2.3005	0.428161966	2.3159	0.428851727
4.4976	0.429472206	4.5017	0.428567296	4.4886	0.429758407	4.4785	0.427784376	4.5069	0.428660498
6.6863	0.429333406	6.6923	0.428181189	6.6812	0.426334186	6.6742	0.427752876	6.6948	0.426943941
8.8803	0.429072823	8.8861	0.427803157	8.8701	0.426294411	8.8606	0.427597303	8.8827	0.426862335
11.0682	0.428645213	11.0712	0.427715402	11.0652	0.425853143	11.06	0.426615287	11.0735	0.426735461
13.2561	0.422834332	13.2632	0.427146136	13.2504	0.425759980	13.24	0.426481862	13.2602	0.426563629

Raw Data Table for the velocity of the 240µm string (Table 3.1)[6]

Qualitative Observation :

1. We note that there was little damping for strings that were relatively thinner than the others, and the pendulum would continue to swing with nearly no damping noticed.
2. The thicker strings, in comparison, exhibit a discernible drop in amplitude, demonstrating that the damping was greater in the thicker than the thinner strings. However, this could've been imagined as we already know through our research, that thicker strings would experience more damping.
3. After every Trial, the string seemed to have dropped down a little. This might be due to the fact that the string loosened over time, near the rod, where it was wound up. Thus the strings had to be rewound after every Trial to ensure that the length of the string is always kept constant, thus preventing any irregularities in the results that have been gathered.

Average Time (s)	Average Velocity (m/s)	θ
2.3233	0.4290199442	0.415309181816608
4.49466	0.4288485566	0.415143271478658
6.68576	0.4277091196	0.414040248986117
8.87596	0.4275260058	0.413862987197975
11.06762	0.4271129012	0.413463084732483
13.25398	0.4257571878	0.412150697673225

Processed Data :

For the processed data, we will use the raw data collected for the string of diameter 240µm as a sample. The steps for the calculation of the processed data are as follows :

[5] For tables of the additional strings, view the appendix.

[6] for rest of the data, view appendix

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1. The first step is to calculate the average time and velocities for each oscillation of the collected data, along with their respective uncertainties.

Average time : $\frac{(2.3069 + 2.3101 + 2.3831 + 2.3005 + 2.3159)}{5} = 2.3233s$

Uncertainty in time : $\frac{(0.0001 + 0.0001 + 0.0001 + 0.0001 + 0.0001)}{5} = \pm 0.0001s$

Therefore, the average time = $2.3233 \pm 0.0001s$

Average velocity :

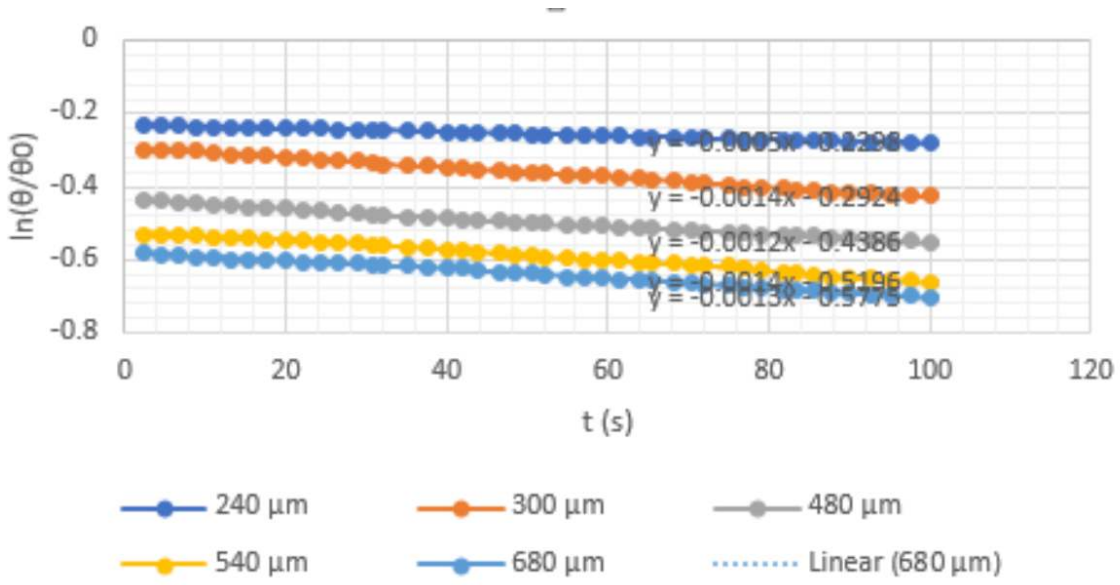
$\frac{(0.429664197 + 0.428593175 + 0.429828656 + 0.428161966 + 0.428851727)}{5} = 0.4290199442ms^{-1}$

Uncertainty in velocity =

$\frac{(0.000000001 + 0.000000001 + 0.000000001 + 0.000000001 + 0.000000001)}{5} = \pm 0.000000001ms^{-1}$

Therefore, average velocity = $0.4290199442 \pm 0.000000001ms^{-1}$

2. The next step is to calculate the value of θ for each oscillation.



$\theta = \frac{vT}{4l}$ (Equation 1.3) $= \frac{0.4290199442 * 2.3233}{4 * 0.6} = 0.4153091818rad$ (0.41531)

Uncertainty of $\theta = \frac{0.0001}{2.3233} + \frac{0.000000001}{0.4290199442} + \frac{0.01}{0.6} = \pm 0.01670971122rad$ (0.01671)

$0.01670971122 * 0.4153091818 = \pm 0.006939696496rad$ (0.00694)

Therefore, $\theta = 0.41531 \pm 0.00694rad$

3. The third step is to calculate θ_0, the inverse sin function of the hypotenuse and the opposite are used to compute in this step using trigonometry. Since the opposite is the displacement of the pendulum from its equilibrium position and the hypotenuse is nothing but the length of the pendulum. The formula for the equation is (with reference to figure 1.1) :

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$$sin^{-1}(\theta_0) = \frac{x_0}{l} = \frac{0.30}{0.60} \quad \rightarrow \quad \theta_0 = sin^{-1}\ \frac{0.30}{0.60} = 0.5235987756rad\ (0.5236)$$

Uncertainty of $\theta_0 = \frac{0.001}{0.30} + \frac{0.00001}{0.60} = 0.00335rad$

$0.00335 * 0.5235987756 = 0.001754055898rad\ (0.00175)$

Therefore, the value of $\theta_0 = 0.5236 \pm 0.00175rad$

Processed Data for θ (Table 3.2)

4. The final step is to substitute the calculated values into the formula (Equation 1.6).

$$\ln\left(\frac{\theta}{\theta_0}\right) = \frac{0.4153091818}{0.5235987756} = -0.2317024364\ (-0.2317)$$

Uncertainty = 0.00335 + 0.006939696496 = 0.0102896965 (0.01029)

0.0102896965 * -0.2317024364 = 0.002384147749 (0.002484)

Therefore the value along with the uncertainty = -0.2317 ± 0.002484

These calculation have been continued for the rest of the strings. The below table shows the variations in $\ln\left(\frac{\theta}{\theta_0}\right)$ with respect to time for the first 13 seconds.

	$\ln\left(\frac{\theta}{\theta_0}\right)$				
Average time (s)	240 µm	300 µm	480 µm	540 µm	680 µm
2.320424	-0.231702436337445	-0.298411773900606	-0.438182504871473	-0.530415519468033	-0.583953738012515
4.49438	-0.232102002476359	-0.30040147880036	-0.439977567169372	-0.533580670940115	-0.585452660385627
6.685736	-0.234762506733723	-0.302133827338562	-0.442674479718693	-0.534665765807858	-0.589202060101449
8.87592	-0.235190725354217	-0.303198401497776	-0.446526978819446	-0.536394812243691	-0.591719132380396
11.067576	-0.236157460261299	-0.307682353242245	-0.44971010963879	-0.538088541642525	-0.596719281393947
13.253932	-0.239336642083611	-0.311242688482311	-0.453297966673001	-0.539039672097823	-0.599159111431307

Processed data - for $\ln\left(\frac{\theta}{\theta_0}\right)$ (Table 3.3)

On graphing the values present in Table 3.2, we get the above graph which is a function of $\ln\left(\frac{\theta}{\theta_0}\right)$ to time graph for which, we know the gradient is $-\frac{k}{2}$.

Processed Data - $\ln\left(\frac{\theta}{\theta_0}\right)$ vs time (Graph 1.1)

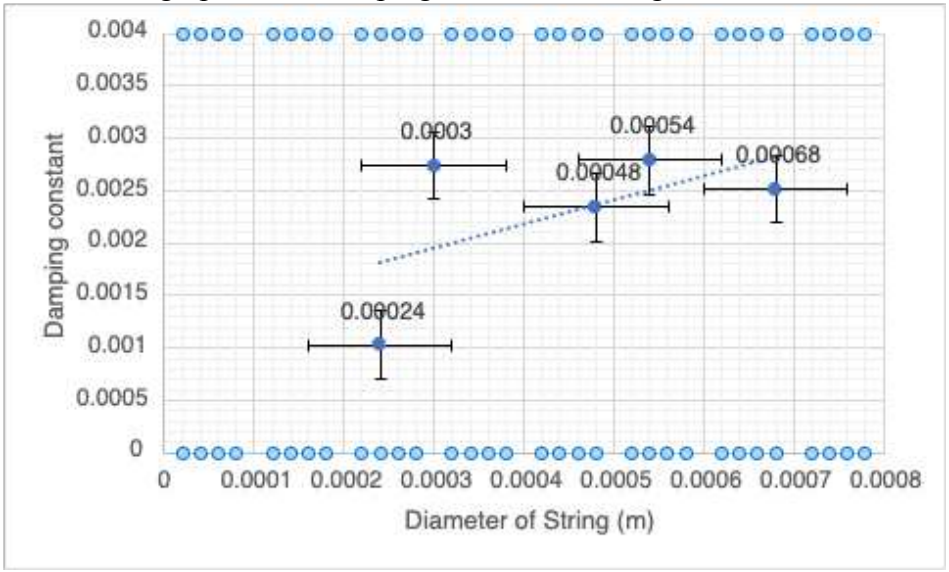
The damping constant(k) of the strings can be calculated by simply multiplying the gradient of the line by -2.

Diameter of the string (µm)	Damping constant of the string ($10^{-3}\ s^{-1}$)
240	1.032
300	2.740
480	2.341
540	2.793
680	2.516

Processed Data for Damping constant of the strings (Table 3.4)

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Using this information, a graph of the damping constant vs string diameter is then made.



Processed Data - Damping constant vs diameter of string (Graph 1.2)

The uncertainty of the damping constants can be calculated by finding the difference between the maximum and minimum gradient and dividing that difference by 2. By doing so we get :

Diameter of the string (μm)	Damping constant of the string (10^{-3} s^{-1})
240	1.032 ± 0.044
300	2.74 ± 0.044
480	2.341 ± 0.044
540	2.793 ± 0.044
680	2.516 ± 0.044

Processed Data (Table 3.4)

4. EVALUATION :

As this was when it was greatest, the velocity at the lowest point of oscillation was determined. As a result, analysing the velocities over 100 seconds would allow one to establish the damping constant. The statistics show that the damping constant typically increases as the diameter of a wire grows, from 0.240 mm, 0.300 mm, 0.480 mm, and 0.540 mm to 0.680 mm.Since the trend line is straight, linear, and proportional, it is likely that the connection between wire diameter and its damping constant is similar.

This is to be expected because air resistance is one of the primary causes of the damping effect. As a result, the surface area of a wire that encounters the air rises along with its diameter. There is greater force opposing the direction of the pendulum when air particles make contact with a bigger surface area. As a result, it is anticipated that the damping effect would grow as air drag increases. The drag force theoretical formula also supports this relationship[7] :

$$FD = \frac{1}{2} 2\rho v^2 C_D A$$

Drag force equation (Equation 2.1)

[7]---. *Damping of a Simple Pendulum Due to Drag on Its String*. 30 Dec. 2016, www.scirp.org/journal/paperinformation.aspx?paperid=73856

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In this equation, the drag force is equal to the cross-sectional area, the velocity of the pendulum relative to the air (v^2), the drag coefficient (C_D), and the density (ρ) of the fluid (in this example, air) (A). Therefore, it is obvious that the drag force grows as the cross-sectional area in contact with the air does. If a result, it is anticipated that the pendulum velocity would decrease more noticeably as the wire's diameter, which affects the cross-sectional area, is raised.

This study was based on an experiment done by Mohazabbi and Shankar that examined how the diameter of the string affected a pendulum's damping constant. In that study, the experiment's results for density, beginning velocity, and drag-coefficient were all the same. Therefore, only the cross-sectional area should have changed, theoretically. The study's findings are as follows :

Diameter of the string (μm)	Damping constant of the string ($10^{-3}\ s^{-1}$)
0.250	0.902
0.279	0.937
0.428	1.075
0.570	1.195
0.718	1.256

Comparison (Table 4.1)

Though the values in the two experiments are different, they display a similar trend. It may be inferred that the results were accurate since the experimental data show a connection that is identical to the theoretical equation. As you can see from the two tables (Table 3.4 and Table 4.1), the string with a diameter of 240μm is the only one whose value comes close to matching the outcomes of Mohazabbi and Shankar's investigation. The other numbers, however, are closer to one another, indicating that they are precise but not accurate. Due to suspicion of shifted apparatus, the experiment had to be reset the next day. The findings of the processed data may have been significantly altered by an undetected flaw in the placement of the apparatus.

The experiment's equipment is very precise, therefore the errors are relatively minimal. However, a few elements had an influence on how the pendulum was dampened. The string had to be tied to the rod that was attached to the clamp stand in order for the experiment to be set up, and in order to secure the string to the rod, I had to wind it around the rod multiple times in addition to securing the string to it. As a result of the string being looped around the rod, there was a substantial amount of surface area of the string in contact with the rod and results in an increase in the friction at the point where the string was suspended when the pendulum is in motion. Thus the damping constant too, increases.

Another thing that I observed was that, after every Trial, the string seemed to have dropped down by a certain amount. Even though I clamp the wound up string to the rod, this kept occurring after every Trial. So, to ensure that the length of the strings were constant during the experiment, the strings had to be rewound to the rod. For the thicker strings, it was harder to wind it around the rod. Winding the excess string seemed to have a significant impact on the damping constant of the pendulum due to the added friction. This can be seen in a few of the thinner strings as their damping constant exceeds that of their consecutive string of a larger diameter.

In addition to this, a number of additional problems were identified throughout the experiment, and they are listed in the table below. The variance in the experimental results might have been caused by these elements (all random). Following are the issues :

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Issue	Effect on the results	Correction	Type of error
The string were coiled when measuring them.	The strings were measured in order to keep their lengths consistent. The strings were coiled while being measured, which might have caused a mistake as different lengths of strings were utilised in the experiment.	It is recommended to use a second person to stretch the string and check for coils before measuring it in order to prevent measurement errors.	Random
The initial displacement of the pendulum is not 30cm.	The initial displacement of the pendulum upon starting the experiment may not always be the same; it may change as a result of human error. Due to changes in the pendulum's damping, this will have an impact on the findings we gathered.	Though, it's nearly never possible to displace the pendulum by exactly 20 cm for every diameter. By measuring 30cm from the setup and setting a ruler at that position, the value of the initial displacement may be made to be close to 30 cm. Make sure the pendulum's bob is correctly positioned in relation to the ruler as you move the pendulum.	Random
Every time the experiment is reset, the equilibrium position of the pendulum may not be perfectly aligned in between the light gate.	The results of the pendulum velocity collected from the light gate through the data logger may vary when the pendulum is displacement.	This may be rectified by securely fastening the clamp stand and the light gate to prevent unintentional equipment displacement.	Random
While in motion, instead on only moving back and forth, the pendulum might swing sideways.	The pendulum may swing in several directions when it is shifted. As a result, the pendulum may encounter more damping. Additionally, the pendulum's bob might collide with the light gate and damage it.	By ensuring the pendulum's string is correctly stretched out and unbent, a mistake of this kind may be prevented. Another technique to prevent this mistake is to make sure that when the pendulum is being displaced, it is properly aligned with the light gate's centre.	Random
Effect of wind on the investigation	Even though the study was being done in a lab on a table, it was just near to the entrance. Many students and lecturers have strolled through the door during the experiment, opening and closing it. This may have allowed air to enter, adding to the string's damping constant by providing additional resistance to the pendulum's swing.	By ensuring that the door remained closed at all times throughout the experiment, this issue might have been prevented. The experiment might also be carried out in an anti-humidity box. This would keep the wind at away and eliminate any humidity that would affect the investigation.	Random

Error Table (Table 5.1)

Conclusion : The other factors are maintained constant in this experiment to demonstrate that any change in the damping constant is caused by the change in the thickness of the string, even though we are aware that the damping constant primarily depends on the mass of the bob, shape of the bob, length of the string, and the drag force on the pendulum. The damping constant is directly related to the string's thickness, meaning that the thicker the string, the higher the damping constant, and the smaller the string, the lower the damping constant, according to the aforementioned experiment.

Extension : This is a really fun experiment and there are many possibilities for the extension of this experiment. Some interesting extensions that have occurred to me at the time of drafting this IA are to find the effect of the mass of the pendulum on the damping constant of the pendulum. It would also be fun to the determine the effect of the material of the string on the damping constant of the pendulum.

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APPENDIX

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Raw Data Tables :

Raw results for String diameter of 240μm :

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)
2.3069	0.429664197	2.3101	0.428593175	2.3831	0.429828656	2.3005	0.428161966	2.3159	0.428851727
4.4976	0.429472206	4.5017	0.428567296	4.4886	0.429758407	4.4785	0.427784376	4.5069	0.428660498
6.6863	0.429333406	6.6923	0.428181189	6.6812	0.426334186	6.6742	0.427752876	6.6948	0.426943941
8.8803	0.429072823	8.8861	0.427803157	8.8701	0.426294411	8.8606	0.427597303	8.8827	0.426862335
11.0682	0.428645213	11.0712	0.427715402	11.0652	0.425853143	11.06	0.426615287	11.0735	0.426735461
13.2561	0.422834332	13.2632	0.427146136	13.2504	0.425759980	13.24	0.426481862	13.2602	0.426563629
15.4458	0.428277539	15.4489	0.426984796	15.4394	0.425095386	15.4288	0.425951868	15.449	0.425575003
17.6415	0.428183893	17.6478	0.426929125	17.6355	0.424409310	17.627	0.425203834	17.6444	0.425342081
19.8295	0.428050134	19.8359	0.426651739	19.8271	0.423580247	19.82	0.425066511	19.8353	0.424981184
22.0294	0.427701171	22.0338	0.426117856	22.0194	0.423166998	22.0136	0.424435591	22.0333	0.424523738
24.2764	0.427560558	24.2847	0.425729805	24.2663	0.423076213	24.2563	0.424269358	24.282	0.424175484
26.4244	0.426950542	26.4278	0.425468787	26.4157	0.422094870	26.4065	0.423197503	26.4325	0.424043583
28.9776	0.426519098	28.9878	0.425270439	28.9715	0.421907538	28.9664	0.422865672	28.9833	0.423946681
30.8261	0.425990624	30.8335	0.424993578	30.8212	0.421525392	30.8122	0.422821905	30.8339	0.423449878
32.1815	0.425941077	32.1854	0.424251351	32.1731	0.421122408	32.1706	0.422427565	32.1869	0.423229676
35.1693	0.425861742	35.1731	0.424246213	35.1654	0.420686411	35.1574	0.422080650	35.1767	0.422138985
37.5879	0.425480352	37.597	0.422918916	37.5834	0.418755459	37.5766	0.421867370	37.598	0.421734314
40.0961	0.425394056	40.0987	0.422797206	40.0856	0.418704794	40.0828	0.420235756	40.1034	0.421354525
41.9834	0.424719721	41.9874	0.421662619	41.9764	0.417466549	41.9685	0.420140310	41.9898	0.419877688
43.8701	0.424613884	43.879	0.421508561	43.8659	0.417176362	43.8573	0.419871740	43.8783	0.419413545
46.4382	0.424590419	46.4438	0.421136408	46.4283	0.416722188	46.4184	0.419629344	46.4453	0.417958532
48.3190	0.422376024	48.3285	0.421092645	48.3156	0.415926682	48.3063	0.418499949	48.3273	0.417437019
50.5157	0.423089691	50.526	0.419545656	50.5116	0.415132282	50.5062	0.418198277	50.5219	0.417319386
52.0898	0.421302067	52.0975	0.419413079	52.0818	0.413861178	52.0729	0.417898256	52.095	0.416784880
54.9758	0.421155536	54.9783	0.419068616	54.9725	0.413613090	54.9666	0.417841870	54.9821	0.416386063
57.0767	0.420772265	57.0791	0.419048452	57.0741	0.413363524	57.0643	0.416675733	57.0839	0.416308149
59.1908	0.420134596	59.1951	0.418802067	59.188	0.412928105	59.1814	0.415801248	59.1977	0.415318146
61.4532	0.419780589	61.462	0.418569951	61.445	0.412292172	61.4414	0.414743848	61.4572	0.415231702
63.8526	0.419416101	63.8631	0.418223086	63.8489	0.412094936	63.8424	0.413849096	63.8596	0.414350090
65.3428	0.419319683	65.3457	0.418117939	65.3405	0.411716719	65.3328	0.412140403	65.3476	0.414271286
68.3391	0.419031067	68.3483	0.417931414	68.3366	0.411267653	68.3327	0.411875935	68.3426	0.414069310
70.3414	0.418530149	70.3441	0.417103023	70.3331	0.411197031	70.325	0.411571360	70.3502	0.413991709
72.0512	0.418337444	72.0609	0.416222548	72.0435	0.408904427	72.038	0.411128725	72.0562	0.413835183
74.9854	0.418024775	74.9929	0.416165490	74.975	0.408084897	74.9688	0.411121239	74.994	0.413497970
76.8421	0.416244127	76.8482	0.416117324	76.8335	0.408031286	76.8291	0.410907103	76.8455	0.413147404
78.9383	0.415599808	78.9452	0.415664682	78.928	0.406724556	78.9257	0.410904375	78.9449	0.412655195
81.9510	0.415220490	81.9589	0.415273888	81.9454	0.405993942	81.937	0.410741049	81.9606	0.412590196
83.3696	0.414825312	83.3802	0.415063456	83.3602	0.405961046	83.3511	0.410653852	83.374	0.412472962
85.3993	0.414391779	85.4052	0.414992371	85.392	0.404308292	85.3817	0.410623800	85.4026	0.412445270
87.7542	0.413822066	87.7578	0.414734175	87.7489	0.403877453	87.7428	0.409790905	87.7569	0.412014476

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89.9856	0.412487398	89.9908	0.414208286	89.976	0.403868358	89.9729	0.409117750	89.9961	0.411849921
92.5889	0.412461873	92.5956	0.414003102	92.5798	0.403846544	92.5761	0.408330772	92.5981	0.411745470
95.0192	0.412347908	95.026	0.413406112	95.0117	0.403583522	95.0043	0.407553100	95.0223	0.411302570
97.4847	0.411737241	97.4923	0.412405758	97.4759	0.403529748	97.4732	0.406917033	97.4922	0.411097837
100.1426	0.409584696	100.1481	0.411829101	100.1382	0.403492574	100.1336	0.405242894	100.1523	0.410972165

Raw results for String diameter of 300μm :

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)
2.3047	0.399759230	2.3001	0.407149079	2.3231	0.404137280	2.3105	0.404440650	2.3059	0.403733959
4.4972	0.399723143	4.4920	0.405364390	4.4983	0.404079310	4.4882	0.403025761	4.4906	0.403013936
6.6857	0.399296185	6.6827	0.404851636	6.6908	0.403986146	6.6838	0.401833554	6.6852	0.401751001
8.8793	0.399073419	8.8766	0.404809244	8.8796	0.403131881	8.8701	0.401026254	8.8732	0.401537240
11.0671	0.395562442	11.0620	0.403740411	11.0744	0.399940492	11.0692	0.400958202	11.0643	0.400385812
13.2549	0.395442390	13.2542	0.402271610	13.2594	0.396073851	13.2490	0.399399217	13.2512	0.400290194
15.4445	0.394862755	15.4403	0.402051487	15.4480	0.394435200	15.4374	0.398975121	15.4404	0.399796346
17.6399	0.394157487	17.6395	0.400478949	17.6438	0.393456430	17.6353	0.398543221	17.6361	0.398282170
19.8272	0.392176388	19.8279	0.400317713	19.8351	0.391756751	19.8280	0.396863086	19.8273	0.397276964
22.0269	0.390910964	22.0260	0.399883544	22.0272	0.391506401	22.0214	0.396822310	22.0255	0.396954944
24.2737	0.390481623	24.2770	0.397897566	24.2740	0.390026522	24.2640	0.394474818	24.2743	0.394783924
26.4216	0.385749036	26.4208	0.397005741	26.4227	0.389731538	26.4135	0.394064371	26.4255	0.394354458
28.9747	0.384406003	28.9812	0.396844018	28.9781	0.388468211	28.9730	0.393351102	28.9767	0.394144022
30.8231	0.383879085	30.8272	0.390574762	30.8275	0.385593729	30.8185	0.392728991	30.8276	0.391498043
32.1783	0.382793402	32.1793	0.389739196	32.1792	0.384641684	32.1767	0.391950681	32.1808	0.391445507
35.1659	0.382742652	35.1671	0.389231834	35.1714	0.380894164	35.1634	0.389237117	35.1707	0.390608423
37.5844	0.382310282	37.5912	0.387224440	37.5892	0.380264405	37.5824	0.389234429	37.5922	0.390548692
40.0924	0.379409889	40.0930	0.387002708	40.0913	0.378594537	40.0885	0.389078799	40.0977	0.390099055
41.9795	0.376632237	41.9879	0.385347192	41.9819	0.378249509	41.9740	0.388930590	41.9843	0.389475973
43.8660	0.374042524	43.8736	0.383924205	43.8713	0.376874222	43.8627	0.388195993	43.8729	0.389264891
46.4340	0.374014582	46.4386	0.383033100	46.4335	0.374812806	46.4236	0.387852583	46.4401	0.385565036
48.3146	0.372714154	48.3235	0.382632298	48.3206	0.373338721	48.3113	0.386647495	48.3223	0.383911843
50.5112	0.372019575	50.5215	0.381823181	50.5161	0.372440895	50.5107	0.385632947	50.5174	0.382184337
52.0849	0.371998806	52.0931	0.379485969	52.0862	0.372210822	52.0773	0.384995076	52.0906	0.382181553
54.9707	0.371471996	54.9740	0.375680801	54.9768	0.371756107	54.9709	0.384319066	54.9778	0.378882089
57.0708	0.371111537	57.0750	0.375122417	57.0782	0.371699598	57.0684	0.383921576	57.0798	0.378372435
59.1848	0.370074187	59.1911	0.374665228	59.1920	0.371526100	59.1854	0.382506569	59.1937	0.377303684
61.4466	0.369415942	61.4581	0.373213914	61.4489	0.370954415	61.4453	0.381667176	61.4533	0.377009373
63.8459	0.368551854	63.8594	0.369376886	63.8526	0.369560391	63.8461	0.380150864	63.8559	0.375880962
65.3359	0.368519214	65.3421	0.368446097	65.3441	0.369393773	65.3364	0.379318495	65.3440	0.375330771
68.3321	0.361444229	68.3448	0.367301870	68.3401	0.368974326	68.3362	0.378979744	68.3391	0.375260662
70.3342	0.360488158	70.3407	0.365750919	70.3365	0.368676706	70.3284	0.377099836	70.3468	0.375172665
72.0438	0.360100319	72.0577	0.365586767	72.0467	0.365611502	72.0412	0.376817835	72.0530	0.373139890
74.9777	0.359950083	74.9898	0.361929722	74.9781	0.362077451	74.9719	0.376185136	74.9909	0.372444709
76.8343	0.359421507	76.8453	0.356552045	76.8364	0.360575469	76.8320	0.376060935	76.8426	0.370120023

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78.9303	0.359398802	78.9424	0.356193241	78.9308	0.359934009	78.9285	0.374663325	78.9421	0.369546258
81.9428	0.359276624	81.9563	0.355748890	81.9480	0.359058254	81.9396	0.370177296	81.9580	0.368297128
83.3611	0.358789964	83.3777	0.355435816	83.3627	0.357450523	83.3536	0.367947693	83.3715	0.367883820
85.3907	0.357317739	85.4028	0.354865605	85.3944	0.357405352	85.3841	0.366499692	85.4002	0.366938103
87.7455	0.357102867	87.7556	0.352753789	87.7511	0.356768208	87.7450	0.365889162	87.7547	0.365773460
89.9766	0.356257993	89.9887	0.351746161	89.9781	0.355980775	89.9750	0.364016840	89.9940	0.364991579
92.5794	0.356059166	92.5937	0.350930261	92.5817	0.355572810	92.5780	0.363468531	92.5962	0.363282555
95.0096	0.353234750	95.0246	0.350611805	95.0131	0.355088413	95.0057	0.362333585	95.0209	0.361676615
97.4748	0.352888277	97.4910	0.350297605	97.4772	0.355057062	97.4745	0.362327177	97.4909	0.361112923
100.1326	0.352683574	100.1471	0.350158220	100.1392	0.353604420	100.1346	0.360737651	100.1513	0.360893122

Raw results for String diameter of 480µm :

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)
2.3201	0.349386950	2.3105	0.349625815	2.3732	0.346676972	2.2959	0.349413065	2.3169	0.349797402
4.5114	0.348576888	4.4882	0.349347419	4.4789	0.346384213	4.4875	0.348507197	4.5073	0.348955092
6.7019	0.348406187	6.6838	0.348471554	6.6716	0.345280847	6.6756	0.346309848	6.6959	0.348611298
8.8956	0.346757808	8.8701	0.348095974	8.8606	0.345188349	8.8637	0.345081790	8.8898	0.345276589
11.0804	0.345983176	11.0692	0.345493542	11.0560	0.344274042	11.0551	0.344858581	11.0774	0.344291835
13.2722	0.344574239	13.2490	0.341789799	13.2414	0.344244804	13.2422	0.344664044	13.2651	0.343450680
15.4575	0.344246687	15.4374	0.341195825	15.4308	0.344075967	15.4318	0.344195826	15.4544	0.342843604
17.6561	0.343423584	17.6353	0.340768786	17.6272	0.343834051	17.6278	0.344021092	17.6498	0.342486930
19.8439	0.342118648	19.8280	0.338279919	19.8191	0.343775115	19.8193	0.342334415	19.8375	0.341612715
22.0416	0.340838484	22.0214	0.337503028	22.0116	0.342026710	22.0177	0.339945148	22.0372	0.341195684
24.2924	0.340617814	24.2640	0.334171853	24.2586	0.340362349	24.2666	0.339576403	24.2841	0.340177808
26.4348	0.339972851	26.4135	0.332733849	26.4087	0.338469351	26.4185	0.338514094	26.4314	0.337598516
28.9944	0.339719676	28.9730	0.330841699	28.9649	0.337340919	28.9701	0.336884669	28.9842	0.337407866
30.8398	0.339712560	30.8185	0.330597745	30.8149	0.334497014	30.8213	0.336222018	30.8324	0.335380708
32.1915	0.338631961	32.1767	0.330138046	32.1670	0.333317484	32.1747	0.335699242	32.1876	0.332997228
35.1791	0.338104372	35.1634	0.330014724	35.1594	0.331517340	35.1647	0.335660323	35.1753	0.331115296
37.6028	0.337895051	37.5824	0.329992328	37.5776	0.330746830	37.5864	0.335376082	37.5937	0.330820263
40.1044	0.336955496	40.0885	0.329639058	40.0799	0.330247837	40.092	0.335092433	40.1018	0.329458851
41.9929	0.335985928	41.9740	0.328035643	41.9709	0.329237426	41.9788	0.334387913	41.9889	0.328893528
43.8844	0.334792631	43.8627	0.328008246	43.8605	0.328473975	43.8675	0.333014067	43.8755	0.328431541
46.449	0.333753059	46.4236	0.327699296	46.4231	0.327529808	46.4349	0.332855998	46.4434	0.326983041
48.3335	0.333597542	48.3113	0.326164508	48.3106	0.327397454	48.3173	0.332580738	48.3240	0.326693187
50.5305	0.331436405	50.5107	0.324575110	50.5071	0.326650378	50.5129	0.332427003	50.5202	0.326654518
52.1019	0.329853951	52.0773	0.324439035	52.0774	0.326228665	52.0862	0.331273533	52.0942	0.326644905
54.9826	0.327655108	54.9709	0.324248955	54.9682	0.325518291	54.9735	0.331038657	54.9801	0.321076048
57.0832	0.327650470	57.0684	0.323481835	57.0700	0.325052770	57.0757	0.330055341	57.0808	0.320799558
59.1991	0.327519348	59.1854	0.322738429	59.1840	0.324663001	59.1897	0.329858157	59.1948	0.320715480
61.4659	0.325832647	61.4453	0.322519384	61.4411	0.324263568	61.4494	0.329074847	61.4571	0.320574866
63.8668	0.325194401	63.8461	0.322478932	63.8452	0.323703352	63.8522	0.327833035	63.8563	0.320292249
65.3493	0.324926632	65.3364	0.322459712	65.3369	0.323143417	65.3404	0.325847720	65.3464	0.319988730

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68.3518	0.323951298	68.3362	0.321021705	68.3331	0.322803872	68.3356	0.323731409	68.3426	0.318378198
70.3475	0.321093724	70.3284	0.320503646	70.3297	0.322173613	70.3434	0.322678887	70.3448	0.318191598
72.0641	0.320667444	72.0412	0.320452713	72.0403	0.321545277	72.0498	0.322019319	72.0544	0.318147564
74.996	0.319111486	74.9721	0.319774285	74.9719	0.321215796	74.9878	0.321351155	74.9885	0.317968270
76.8511	0.318382576	76.8320	0.319682232	76.8306	0.320025311	76.8397	0.318466339	76.845	0.317872273
78.948	0.318309285	78.9285	0.319114773	78.9252	0.319899715	78.9393	0.315018314	78.9411	0.317452494
81.9615	0.317574873	81.9396	0.318290171	81.9428	0.319287538	81.9554	0.314390797	81.9536	0.317114523
83.3827	0.317046209	83.3536	0.317902225	83.3577	0.318822437	83.369	0.314378634	83.3721	0.316630306
85.4076	0.316983869	85.3841	0.317655621	85.3896	0.317960662	85.3978	0.313698755	85.4017	0.315937168
87.7600	0.313713827	87.7450	0.316899489	87.7467	0.317229737	87.7525	0.313208001	87.7564	0.315139596
89.9929	0.313337785	89.9750	0.315366097	89.9739	0.315739595	89.9919	0.312898299	89.9877	0.313635825
92.5975	0.311958998	92.5780	0.314734244	92.5779	0.315483431	92.5943	0.312070823	92.5908	0.313270594
95.0274	0.311602417	95.0057	0.312614924	95.0103	0.313822125	95.0195	0.311138267	95.0206	0.313004052
97.4936	0.311405950	97.4745	0.312007048	97.4746	0.313804498	97.4896	0.310454204	97.4860	0.312872553
100.1491	0.310132339	100.1346	0.310763537	100.1372	0.311893788	100.1503	0.308803386	100.1436	0.311771016

Raw results for String diameter of 540µm :

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)
2.3138	0.318962529	2.317	0.317860393	2.3701	0.317463842	2.3074	0.319299166	2.2928	0.319701644
4.5041	0.318764161	4.5082	0.317014684	4.4757	0.316663036	4.485	0.318559071	4.4843	0.317251598
6.6926	0.317952186	6.6986	0.316577503	6.6683	0.316660707	6.6805	0.318126448	6.6723	0.317213236
8.8864	0.317788300	8.8922	0.315513243	8.8572	0.315635609	8.8667	0.317649401	8.8603	0.317202713
11.0739	0.317738308	11.0769	0.314649641	11.0525	0.314802711	11.0657	0.316717392	11.0516	0.317200974
13.2614	0.317256746	13.2685	0.314374910	13.2377	0.314699972	13.2453	0.316695459	13.2385	0.316578813
15.4506	0.316864190	15.4537	0.312075119	15.427	0.314088218	15.4336	0.315874097	15.428	0.316408084
17.6458	0.316861036	17.6521	0.312016194	17.6232	0.313940983	17.6313	0.312561417	17.6238	0.315737724
19.8333	0.316070907	19.8397	0.311473369	19.8149	0.312725080	19.8238	0.312553699	19.8151	0.315422589
22.0328	0.315486720	22.0372	0.311194746	22.0072	0.312347670	22.017	0.309667668	22.0133	0.315408638
24.2796	0.315321758	24.2879	0.310982039	24.2541	0.311119648	24.2595	0.309135695	24.2621	0.314709814
26.4268	0.314075196	26.4302	0.309695152	26.4041	0.310544299	26.4089	0.308263976	26.4139	0.314607640
28.9795	0.313109966	28.9897	0.307838964	28.9602	0.309178315	28.9683	0.307851341	28.9654	0.313717301
30.8274	0.311957521	30.8348	0.306492780	30.8099	0.308335322	30.8135	0.306250803	30.8163	0.313470324
32.1824	0.311867997	32.1863	0.306397392	32.1618	0.306213187	32.1715	0.305025069	32.1695	0.313326527
35.17	0.311438361	35.1738	0.305752510	35.1541	0.306022683	35.1581	0.304927287	35.1594	0.312268628
37.5883	0.310363254	37.5974	0.302896969	37.5722	0.306005669	37.577	0.299973169	37.581	0.311634675
40.0963	0.309788824	40.0989	0.302769419	40.0744	0.305801871	40.083	0.297611278	40.0865	0.309738921
41.9832	0.308521386	41.9872	0.302376041	41.9652	0.305395450	41.9683	0.297170137	41.9731	0.308463331
43.8696	0.308303904	43.8785	0.302024827	43.8546	0.305366467	43.8568	0.296259250	43.8616	0.307508259
46.4374	0.307335924	46.443	0.301651208	46.4171	0.297534085	46.4176	0.295690523	46.4289	0.307335459
48.3178	0.307204776	48.3273	0.301321700	48.3044	0.296997456	48.3051	0.295445470	48.3111	0.306676992
50.5138	0.306919854	50.5241	0.300418702	50.5007	0.296538000	50.5043	0.294160380	50.5065	0.305724866
52.0877	0.306213718	52.0954	0.299841347	52.0709	0.295032779	52.0708	0.293379562	52.0797	0.303926096
54.9735	0.305164458	54.976	0.298187727	54.9616	0.293304725	54.9643	0.292825902	54.9669	0.303425429

Clastify

57.0741	0.304264894	57.0765	0.297147613	57.0633	0.292725171	57.0617	0.291211418	57.069	0.302430491
59.188	0.304131083	59.1923	0.296462254	59.1772	0.291201686	59.1786	0.291193652	59.1829	0.301841995
61.4502	0.303678409	61.459	0.296100471	61.4342	0.290910901	61.4384	0.290614734	61.4425	0.300485626
63.8493	0.303190658	63.8598	0.295536606	63.8382	0.289020682	63.8391	0.290571052	63.8452	0.299609638
65.3393	0.302980711	65.3422	0.292878132	65.3298	0.288216489	65.3293	0.288708964	65.3333	0.298592702
68.3354	0.301440361	68.3446	0.292765672	68.3259	0.288140935	68.329	0.288620664	68.3284	0.298563147
70.3375	0.298753904	70.3402	0.292437846	70.3224	0.287738653	70.3211	0.288107183	70.3361	0.298359404
72.047	0.298177385	72.0567	0.291992633	72.0329	0.287668424	72.0338	0.288101568	72.0424	0.297391427
74.981	0.297906019	74.9885	0.290568944	74.9698	0.286031312	74.9644	0.287101782	74.9803	0.297278939
76.8372	0.297265004	76.8433	0.289772195	76.8228	0.284768350	76.8242	0.281666765	76.8319	0.296682901
78.9332	0.294101281	78.9401	0.285445611	78.9173	0.284596768	78.9206	0.281313282	78.9314	0.296506551
81.9456	0.293672470	81.9535	0.284545988	81.9348	0.283597282	81.9316	0.280838467	81.9474	0.295573657
83.364	0.291404519	83.3746	0.284318315	83.3496	0.278984552	83.3455	0.280764785	83.3609	0.295384887
85.3935	0.288836113	85.3994	0.282707576	85.3814	0.276414744	85.3759	0.277646626	85.3896	0.295362196
87.7481	0.288601399	87.7517	0.282278559	87.7384	0.275076906	87.7367	0.277188447	87.7442	0.295241881
89.9793	0.287504748	89.9845	0.282005443	89.9655	0.274540577	89.9666	0.275263188	89.9835	0.293764408
92.5822	0.286903423	92.5889	0.281257111	92.5693	0.274347185	92.5694	0.274380739	92.5857	0.292590291
95.0118	0.286142492	95.0186	0.281245030	95.0015	0.273401499	94.9969	0.274102445	95.0107	0.291682846
97.4771	0.283874327	97.4847	0.280566238	97.4657	0.273112960	97.4656	0.271745868	97.4807	0.290780829
100.1346	0.280617492	100.1401	0.280201443	100.1282	0.271616026	100.1256	0.270296377	100.1413	0.290617649

Raw results for String diameter of 680µm :

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)	Time (s)	Velocity (m/s)
2.32	0.299270134	2.3232	0.299816659	2.3763	0.298869749	2.3136	0.307993443	2.299	0.300254233
4.5105	0.299122453	4.5146	0.298980051	4.4821	0.298797047	4.4914	0.306815894	4.4907	0.300232781
6.6992	0.298244140	6.7052	0.298283787	6.6749	0.296082860	6.6871	0.306311037	6.6789	0.299398057
8.8932	0.297415444	8.899	0.297551095	8.864	0.294897339	8.8735	0.305981221	8.8671	0.298708145
11.0809	0.296827870	11.0839	0.297510943	11.0595	0.293512283	11.0727	0.301433180	11.0586	0.297814631
13.2688	0.295720262	13.2759	0.295761502	13.2451	0.293503881	13.2527	0.301046676	13.2459	0.297442740
15.4582	0.295234320	15.4613	0.295580018	15.4346	0.293056867	15.4412	0.300299939	15.4356	0.297170785
17.6538	0.295222898	17.6601	0.295221069	17.6312	0.292866357	17.6393	0.299659137	17.6318	0.296877935
19.8417	0.295191281	19.8481	0.294585535	19.8233	0.292742230	19.8322	0.299500393	19.8235	0.296466002
22.0416	0.292380450	22.046	0.294471852	22.016	0.291581996	22.0258	0.299446774	22.0221	0.295318066
24.2886	0.292048808	24.2969	0.294184749	24.2631	0.291172570	24.2685	0.298678082	24.2711	0.295255983
26.436	0.291935769	26.4394	0.293528163	26.4133	0.291160365	26.4181	0.297493408	26.4231	0.294782027
28.9889	0.291831007	28.9991	0.293293880	28.9696	0.290996238	28.9777	0.295733940	28.9748	0.294730940
30.8374	0.290931149	30.8448	0.293046008	30.8199	0.290856497	30.8235	0.295408904	30.8263	0.293513442
32.1928	0.288979533	32.1967	0.291423138	32.1722	0.290792574	32.1819	0.295258386	32.1799	0.292745753
35.1806	0.287189696	35.1844	0.291331970	35.1647	0.289751684	35.1687	0.294650204	35.17	0.292182951
37.5991	0.286509780	37.6082	0.290541297	37.583	0.289659134	37.5878	0.293534099	37.5918	0.290896466
40.1073	0.285923383	40.1099	0.289432434	40.0854	0.289179173	40.094	0.292973104	40.0975	0.290729125
41.9946	0.285165244	41.9986	0.289290370	41.9766	0.289096951	41.9797	0.292599561	41.9845	0.289904565
43.8814	0.284937424	43.8903	0.283498843	43.8664	0.289001859	43.8686	0.291779084	43.8734	0.288445697

Clastify

46.4494	0.283965509	46.455	0.283343813	46.4291	0.288853459	46.4296	0.290773630	46.4409	0.287270018
48.3302	0.282693211	48.3397	0.282655856	48.3168	0.288653280	48.3175	0.290579433	48.3235	0.286555025
50.5266	0.279938604	50.5369	0.281868419	50.5135	0.288483662	50.5171	0.290524139	50.5193	0.286034172
52.1007	0.278384284	52.1084	0.280498801	52.0839	0.288333176	52.0838	0.289001390	52.0927	0.285780813
54.9867	0.278044585	54.9892	0.278342467	54.9748	0.285867207	54.9775	0.288755040	54.9801	0.283879919
57.0875	0.277940395	57.0899	0.278008821	57.0767	0.285502754	57.0751	0.287272765	57.0824	0.283460853
59.2016	0.277619358	59.2059	0.277645704	59.1908	0.285475942	59.1922	0.286536023	59.1965	0.283131655
61.464	0.277550422	61.4728	0.274954162	61.448	0.282839038	61.4522	0.286096084	61.4563	0.281859217
63.8633	0.277550052	63.8738	0.273381726	63.8522	0.282797816	63.8531	0.285971051	63.8592	0.281463943
65.3535	0.277433503	65.3564	0.273177060	65.344	0.281598842	65.3435	0.284683398	65.3475	0.280303182
68.3498	0.276984166	68.359	0.273165472	68.3403	0.281402588	68.3434	0.284359213	68.3428	0.279385312
70.3521	0.270824289	70.3548	0.272945338	70.337	0.280409461	70.3357	0.283985944	70.3507	0.278990871
72.0618	0.268278372	72.0715	0.272042583	72.0477	0.279644814	72.0486	0.283309730	72.0572	0.278950992
74.996	0.267987415	75.0035	0.271820308	74.974	0.278571099	74.9794	0.281881456	74.9953	0.278936742
76.8528	0.267977390	76.8589	0.268905124	76.8384	0.278071885	76.8398	0.281317163	76.8475	0.277824990
78.949	0.266889978	78.9559	0.268255357	78.9331	0.276615715	78.9364	0.280904727	78.9472	0.276174372
81.9616	0.266366238	81.9695	0.266877343	81.9508	0.276357287	81.9476	0.279805375	81.9634	0.275485014
83.3802	0.266100642	83.3908	0.266276149	83.3658	0.276281255	83.3617	0.278545420	83.3771	0.275263695
85.4099	0.264665002	85.4158	0.265458635	85.3978	0.276227561	85.3923	0.278522459	85.406	0.274534719
87.7647	0.264545082	87.7683	0.264662725	87.755	0.275372872	87.7533	0.276503274	87.7608	0.273956840
89.9961	0.262015184	90.0013	0.264479858	89.9823	0.273568801	89.9834	0.275747702	90.0003	0.273950774
92.5994	0.261431662	92.6061	0.264127416	92.5865	0.272839120	92.5866	0.275398810	92.6029	0.271354037
95.0294	0.261243128	95.0362	0.264071654	95.0191	0.271992914	95.0145	0.275171776	95.0283	0.271032349
97.4949	0.260679219	97.5025	0.263005522	97.4835	0.271621263	97.4834	0.275077137	97.4985	0.270726492
100.1526	0.260378585	100.1581	0.261499765	100.1462	0.270440322	100.1436	0.271337210	100.1593	0.270229008

Processed Data Tables :

Processed Data for string of diameter 240μm :

Average Time (s)	Average Velocity (m/s)	θ	$\ln\left(\frac{\theta}{\theta_0}\right)$
2.3233	0.4290199442	0.415309181816608	-0.231702436337445
4.49466	0.4288485566	0.415143271478658	-0.232102002476359
6.68576	0.4277091196	0.414040248986117	-0.234762506733723
8.87596	0.4275260058	0.413862987197975	-0.235190725354217
11.06762	0.4271129012	0.413463084732483	-0.236157460261299
13.25398	0.4257571878	0.412150697673225	-0.239336642083611
15.44238	0.4263769184	0.412750622716133	-0.237882104107128
17.63924	0.4260136486	0.412398962413492	-0.238734459438255
19.82956	0.425665963	0.412062388265792	-0.239550929891649
22.0259	0.4251890708	0.411600736745683	-0.24067190169294
24.27314	0.4249622836	0.411381197286617	-0.24120542364572
26.42138	0.424351057	0.410789504470042	-0.242644766880554
28.97732	0.4241018856	0.4105482961727	-0.243232121569667

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30.82538	0.4237562754	0.410213731098675	-0.244047376372981
32.1795	0.4233944154	0.409863435541175	-0.244901675439104
35.16838	0.4230028002	0.409484335710275	-0.245827045279003
37.58858	0.4221512822	0.408660030806358	-0.247842105741617
40.09332	0.4216972674	0.408220525562675	-0.248918163392065
41.9811	0.4207733774	0.407326161547258	-0.251111451374426
43.87012	0.4205168184	0.4070778017453	-0.251721369353344
46.4348	0.4200073782	0.406584642405025	-0.252933565895555
48.31934	0.4190664638	0.405673798061058	-0.255176311997973
50.51628	0.4186570584	0.4052774765753	-0.256153735746368
52.0874	0.417851892	0.4044980419515	-0.258078799783683
54.97506	0.417613035	0.404266818423125	-0.258650593999786
57.07562	0.4172336246	0.403899533347158	-0.259559528405306
59.1906	0.4165968324	0.40328309196455	-0.261086918820114
61.45176	0.4161236524	0.402825034008717	-0.262223386721565
63.85332	0.4155866618	0.402305204733308	-0.263514679291704
65.34188	0.415113206	0.401846879791583	-0.264654575587871
68.33986	0.4148350758	0.401577638169225	-0.265324810629043
70.33876	0.4144786544	0.401232607403133	-0.266184368137778
72.04996	0.4136856654	0.400464961009925	-0.268099421040669
74.98322	0.4133788742	0.400167974345358	-0.268841300783742
76.83968	0.4128894488	0.399694190165433	-0.270025965483128
78.93642	0.4123097232	0.3991329916294	-0.271431021898704
81.95058	0.411963913	0.398798232947042	-0.272270088454035
83.36702	0.4117953256	0.398635033319367	-0.272679400773591
85.39616	0.4113523024	0.398206168402467	-0.273755813382233
87.75212	0.410847815	0.397717803578958	-0.274982978041139
89.98428	0.4103063426	0.397193635734408	-0.276301786387827
92.5877	0.4100775522	0.396972157094275	-0.276859550642784
95.0167	0.4096386424	0.396547274119967	-0.277930433089259
97.48366	0.4091375234	0.396062170048008	-0.27915450160396
100.14296	0.408224286	0.39517811819325	-0.28138910021014

Processed Data for string of diameter 300µm :

Average Time (s)	Average Velocity (m/s)	θ	$\ln\left(\frac{\theta}{\theta_0}\right)$
2.30886	0.4038440396	0.38850806219619	-0.298411773900606
4.49326	0.403041308	0.3877358143287	-0.30040147880036
6.68564	0.4023437044	0.38706470222541	-0.302133827338562
8.87576	0.4019156076	0.38665286240139	-0.303198401497776
11.0674	0.4001174718	0.384923010808395	-0.307682353242245
13.25374	0.3986954524	0.38355499259511	-0.311242688482311
15.44212	0.3980241818	0.382909213496145	-0.312927775004659
17.63892	0.3969836514	0.381908197238085	-0.315545437248233

Clastify

19.8291	0.3956781804	0.38065230149931	-0.3188393316379
22.0254	0.3952156326	0.380207318952015	-0.320009015469367
24.2726	0.3935328906	0.378588479079465	-0.324275887618173
26.42082	0.3921810288	0.37728795423132	-0.327716995381688
28.97674	0.3914426712	0.37657763576118	-0.329601465742925
30.82478	0.388854922	0.37408815633705	-0.336234213836789
32.17886	0.388114094	0.37337546128035	-0.338141183636738
35.1677	0.386542838	0.37186387372695	-0.342197839144145
37.58788	0.3859164496	0.37126127242644	-0.343819642512559
40.09258	0.3848369976	0.37022281261614	-0.34662067495603
41.98152	0.3837271002	0.369155063569905	-0.349508913359914
43.8693	0.382460367	0.367936434563175	-0.352815504523004
46.43396	0.3810556214	0.366585034177335	-0.356495184336834
48.31846	0.3798489022	0.365424140138955	-0.359666988877982
50.51538	0.378820187	0.364434490398675	-0.362378884868231
52.08642	0.3781744452	0.36381327064353	-0.364084952333337
54.97404	0.3764220118	0.362127385901895	-0.368729650924783
57.07444	0.3760455126	0.361765184259015	-0.369730356521265
59.1894	0.3752151536	0.36096635814204	-0.371940932358649
61.45044	0.374452164	0.3602323430721	-0.373976474913425
63.85198	0.3727041914	0.358550749731585	-0.378655484270474
65.3405	0.37220167	0.35806731158175	-0.380004705701725
68.33846	0.3703921662	0.356326523688555	-0.38487818375451
70.33732	0.3694376568	0.35540826178302	-0.387458533721315
72.04848	0.3682512626	0.354266920902765	-0.390675052748967
74.98168	0.3665174202	0.352598921167905	-0.395394485296238
76.83812	0.3645459958	0.350702361609495	-0.400787804495547
78.93482	0.363947127	0.350126234852175	-0.40243193514321
81.94894	0.3625116384	0.34874525893176	-0.406383957024464
83.36532	0.3615015632	0.34777354133748	-0.409174170940671
85.39444	0.3606052982	0.346911312000855	-0.411656533339233
87.75038	0.3596574972	0.34599950374383	-0.414288354816423
89.98248	0.3585986696	0.34498088512194	-0.417236685553962
92.5858	0.3578626646	0.344272829911815	-0.419291242370274
95.01478	0.3565890336	0.34304756504904	-0.422856584442726
97.48168	0.3563366088	0.34280472608082	-0.423564722277967
100.14096	0.3556153974	0.342110902683735	-0.425590734257475

Processed Data for string of diameter 480μm :

Average Time (s)	Average Velocity (m/s)	θ	$\ln\left(\frac{\theta}{\theta_0}\right)$
2.32332	0.3489800408	0.33783012849644	-0.438182504871473
4.49466	0.3483541618	0.33722424633049	-0.439977567169372
6.68576	0.3474159468	0.33631600729974	-0.442674479718693

Clastify

8.87596	0.346080102	0.3350228427411	-0.446526978819446
11.06762	0.3449802352	0.33395811668536	-0.44971010963879
13.25398	0.3437447132	0.33276206961326	-0.453297966673001
15.44238	0.3433115818	0.33234277676149	-0.454558799185653
17.63924	0.3429068886	0.33195101350923	-0.45573828724776
19.82956	0.3416241624	0.33070927041132	-0.459486043319664
22.0259	0.3403018108	0.32942916794494	-0.463364333073804
24.27314	0.3389812454	0.32815079460947	-0.467252453207469
26.42138	0.3374577322	0.32667595765621	-0.471756971209959
28.97732	0.3364389658	0.32568974084269	-0.474780482614961
30.82538	0.335282009	0.32456974881245	-0.478225240148149
32.1795	0.3341567922	0.32348048268921	-0.48158691520386
35.16838	0.333282411	0.32263403796855	-0.484207024109733
37.58858	0.3329661108	0.32232784355994	-0.485156520321473
40.09332	0.332278735	0.32166242941675	-0.487223055820285
41.9811	0.3313080876	0.32072279420118	-0.490148515034365
43.87012	0.330544092	0.3199832082606	-0.492457175375402
46.4348	0.3297642404	0.31922827291922	-0.494819259645216
48.31934	0.3292866858	0.31876597618869	-0.496268479013847
50.51628	0.3283486828	0.31785794238454	-0.499121134697775
52.0874	0.3276880178	0.31721838563129	-0.501135245258163
54.97506	0.3259074118	0.31549466999299	-0.506583907609719
57.07562	0.3254079948	0.31501120936614	-0.508117472124906
59.1906	0.325098883	0.31471197368815	-0.509067844315624
61.45176	0.3244530624	0.31408678705632	-0.511056355988574
63.85332	0.3239003938	0.31355177621809	-0.512761193769801
65.34188	0.3232732422	0.31294466211171	-0.514699319062395
68.33986	0.3219772964	0.31169012178002	-0.518716200769631
70.33876	0.3209282936	0.31067463461948	-0.521979522001371
72.04996	0.3205664634	0.31032436489437	-0.523107606789042
74.98326	0.3198841984	0.30966389826112	-0.52523818545429
76.83968	0.3188857462	0.30869734660891	-0.52836435950676
78.93642	0.3179589162	0.30780012882741	-0.531275055659447
81.95058	0.3173315804	0.30719283640622	-0.533250013595275
83.36702	0.3169559622	0.30682921920771	-0.534434392031492
85.39616	0.316447215	0.30633672648075	-0.53604078532952
87.75212	0.31523813	0.3051662717465	-0.539868914274166
89.98428	0.3141955202	0.30415697332961	-0.543181767831398
92.5877	0.313503618	0.3034871774049	-0.545386335077455
95.0167	0.312436357	0.30245401539385	-0.548796444918263
97.48366	0.3121088506	0.30213697282333	-0.549845228660527
100.14296	0.3106728132	0.30074681681826	-0.554456924953392

Clastify

Processed Data for string of diameter 540µm :

Average Time (s)	Average Velocity (m/s)	θ	$\ln\left(\frac{\theta}{\theta_0}\right)$
2.32022	0.3186575148	0.30806480791219	-0.530415519468033
4.49146	0.31765051	0.307091277630083	-0.533580670940115
6.68246	0.317306016	0.3067582351848	-0.534665765807858
8.87256	0.3167578532	0.306228294229877	-0.536394812243691
11.06412	0.3162218052	0.30571006535881	-0.538088541642525
13.25028	0.31592118	0.3054194334415	-0.539039672097823
15.43858	0.3150619416	0.30458875755798	-0.541763164697534
17.63524	0.3142234708	0.303778158924823	-0.544428000969496
19.82536	0.3136491288	0.30322290901014	-0.546257487271527
22.0215	0.3128210884	0.302422394053103	-0.548900999634836
24.26864	0.3122537908	0.301873954370823	-0.550716134987846
26.41678	0.3114372526	0.301084559261488	-0.553334542531645
28.97262	0.3103391774	0.300022985911262	-0.556866604181463
30.82038	0.30930135	0.29901965762375	-0.560216379816087
32.1743	0.3085660344	0.29830878513982	-0.562596553833067
35.16308	0.3080818938	0.297840738180265	-0.564166787599649
37.58318	0.3061747472	0.295996988311827	-0.570376415956639
40.08782	0.3051420626	0.294998631869072	-0.573754977009625
41.9754	0.304385269	0.294266995349658	-0.576238192936637
43.86422	0.3038925414	0.293790646836295	-0.57785826751225
46.4288	0.3019094398	0.291873466838648	-0.584405319998571
48.31314	0.3015292788	0.29150594302389	-0.58566530229197
50.50988	0.3007523604	0.29075485068637	-0.588245220989015
52.0809	0.2996787004	0.289716880934203	-0.591821522131417
54.96846	0.2985816482	0.288656296577752	-0.595489000453239
57.06892	0.2975559174	0.287664662779095	-0.598930259221139
59.1838	0.296966134	0.287094484762283	-0.600914318826951
61.44486	0.2963580282	0.286506593412585	-0.602964146060255
63.84632	0.2955857272	0.285759964984993	-0.605573520451524
65.33478	0.2942753996	0.284493194858297	-0.610016361980917
68.33266	0.2939061558	0.284136225337615	-0.611271905767071
70.33146	0.293079398	0.283336950344817	-0.614088868949079
72.04256	0.2926662874	0.282937572229678	-0.615499415125663
74.9768	0.2917773992	0.282078232154927	-0.618541244153047
76.83188	0.290031043	0.280389927745608	-0.624544462112841
78.92852	0.2883926986	0.278806044644038	-0.630209336022686
81.94258	0.2876455728	0.27808375455084	-0.632803351927669
83.35892	0.2861714116	0.276658596926063	-0.637941451646706
85.38796	0.284193451	0.274746387033008	-0.644877252425812
87.74382	0.2836774384	0.274247527551853	-0.646694611799767
89.97588	0.2826156728	0.273221056810007	-0.650444496172999
92.5791	0.2818957498	0.272525065250398	-0.652995103163302

Clastify

95.0079	0.2813148624	0.27196348751572	-0.655057875395895
97.47476	0.2800160444	0.270707844390737	-0.659685521235899
100.13396	0.2786697974	0.269406348884762	-0.664504865149577

Processed Data for string of diameter 680μm :

Average Time (s)	Average Velocity (m/s)	θ	$\ln\left(\frac{\theta}{\theta_0}\right)$
2.32642	0.3012408436	0.292005301403297	-0.583953738012515
4.49786	0.3007896452	0.291567935994243	-0.585452660385627
6.68906	0.2996639762	0.290476778129668	-0.589202060101449
8.87936	0.2989106488	0.289746546492207	-0.591719132380396
11.07112	0.2974197814	0.288301386601912	-0.596719281393947
13.25768	0.2966950122	0.287598837617635	-0.599159111431307
15.44618	0.2962683858	0.287185290872015	-0.600598075370916
17.64324	0.2959694792	0.286895548250193	-0.601607489469693
19.83376	0.2956970882	0.286631508304268	-0.602528247999084
22.0303	0.2946398276	0.285606661552163	-0.606110140650844
24.27764	0.2942680384	0.285246270789387	-0.607372780431558
26.42598	0.2937799464	0.28477314287662	-0.609032822212856
28.98202	0.293317201	0.284324584479342	-0.61060920696579
30.83038	0.2927512	0.283775936126667	-0.612540726136598
32.1847	0.2918398768	0.282892552577107	-0.615658542929901
35.17368	0.291021301	0.282099072946842	-0.618467363861387
37.59398	0.2902281552	0.28133024367516	-0.621196471936487
40.09882	0.2896474438	0.280767335918832	-0.623199355296746
41.9868	0.2892113382	0.280344600589685	-0.624706132756367
43.87602	0.2875325814	0.278717311675245	-0.630527647241531
46.4408	0.2868412858	0.278047210046182	-0.632934776023923
48.32554	0.286227361	0.277452107157342	-0.63507736413545
50.52268	0.2853697992	0.27662083677286	-0.638077947424288
52.0939	0.2843996928	0.27568047221824	-0.64148320990693
54.98166	0.2829778436	0.274302214544963	-0.646495224254411
57.08232	0.2824371176	0.273778066302913	-0.648407894423936
59.1974	0.2820817364	0.273433580498203	-0.649666953335533
61.45866	0.2806597846	0.272055223370472	-0.654720622820622
63.86032	0.2802329176	0.271641443401247	-0.656242721720248
65.34898	0.279439197	0.270872056951975	-0.65907910077656
68.34706	0.2790593502	0.270503855621785	-0.660439343736806
70.34606	0.2774311806	0.268925602988105	-0.666290923074554
72.05736	0.2764452982	0.267969946099352	-0.669850862804522
74.98964	0.275839404	0.2673826276057	-0.672045000866516
76.84748	0.2748193104	0.26639380837532	-0.675749999044977
78.94432	0.2737680298	0.265374758286382	-0.679582686421665
81.95858	0.2729782514	0.264609193175828	-0.682471700856604

Clastify

83.37512	0.2724934322	0.264139237724468	-0.684249315872166
85.40436	0.2718816752	0.263546236174493	-0.686496873665346
87.76042	0.2710081586	0.262699500137588	-0.689714901705064
89.99268	0.2699524638	0.261676171180665	-0.693617944168663
92.5963	0.269030209	0.260782191175742	-0.697040153254873
95.0255	0.2687023642	0.260464397550902	-0.698259513423702
97.49256	0.2682219266	0.259998689366988	-0.700049105500109
100.15196	0.266776978	0.258598040482817	-0.705450806703649